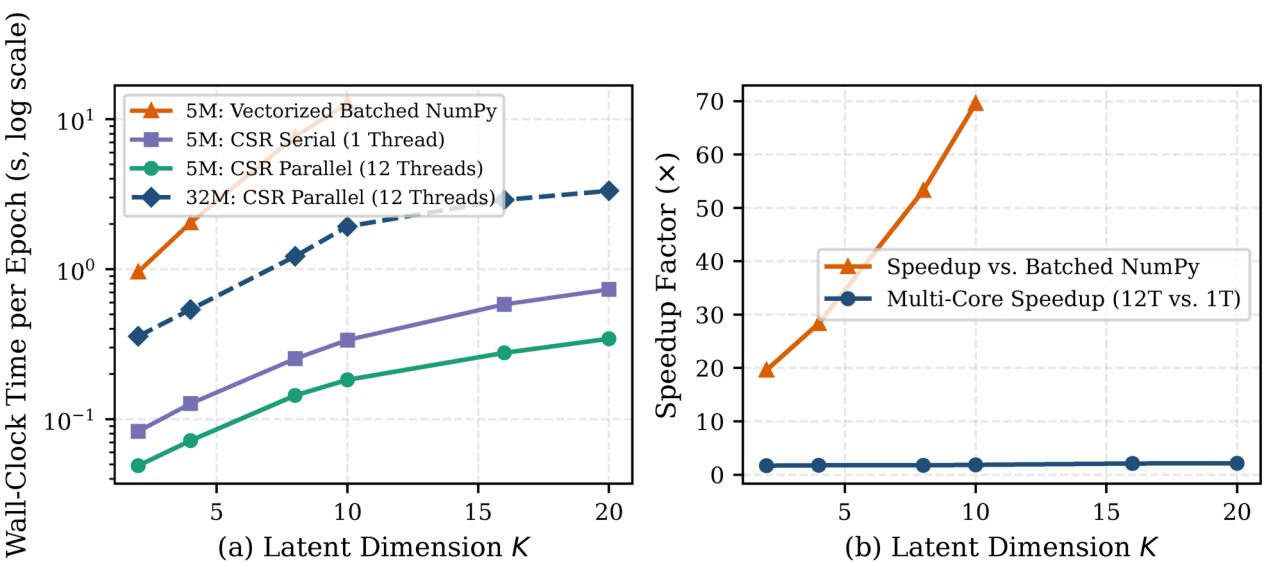

Wall-Clock Time per Epoch (s, log scale)
5M: Vectorized Batched NumPy
5M: CSR Serial (1 Thread)
5M: CSR Parallel (12 Threads)
32M: CSR Parallel (12 Threads)
(a) Latent Dimension K
Speedup Factor (×)
Speedup vs. Batched NumPy
Multi-Core Speedup (12T vs. 1T)
(b) Latent Dimension K